Informatica® Cloud Application Integration
October 2024

Update Amazon Bedrock Knowledge Base

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Preface

Use *Update Amazon Bedrock Knowledge Base* to learn how to update the existing Amazon Bedrock knowledge base using Amazon S3 data source and provide notification status. This guide assumes that you have an understanding of the Amazon Bedrock Connector, Amazon S3 Connector, and Email Connector concepts.

CHAPTER 1

Introduction to Update Amazon Bedrock Knowledge Base recipe

The Update Amazon Bedrock Knowledge Base recipe is based on REST and SOAP APIs. Use the recipe to update the existing Amazon Bedrock knowledge base using Amazon S3 data source and provide notification status.

The process is triggered by an HTTP request with the required input parameters, that is, knowledge base ID, email address, and an additional parameter data source ID if it is present or the bucket name to create a new data source using the bucket name.

The process then creates a data source if the user doesn't provide a data source ID. The name of the data source is the same as the name of the bucket to be added. If the user provides data source ID, the process updates the vector database.

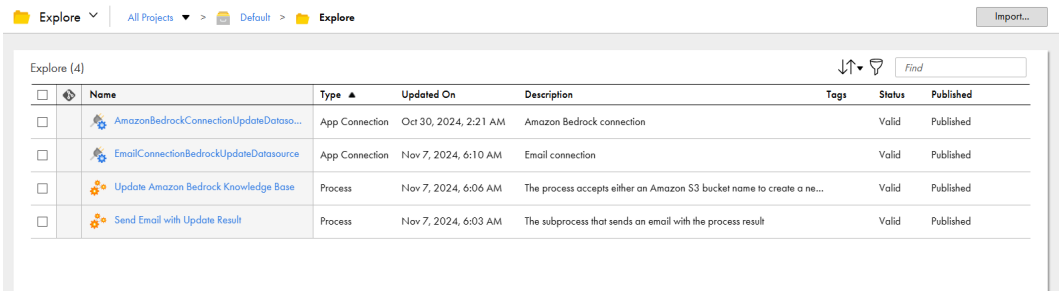
The process then starts an ingestion job splitting the dataset into chunks and upserting them into the vector database. The process checks the ingestion status every 30 seconds, by default. You can change the interval as required.

The process then sends an email notification when the processing of files in the bucket is complete or to report any errors encountered during the process.

Update Amazon Bedrock Knowledge Base recipe contents

The Update Amazon Bedrock Knowledge Base recipe contains app connections and processes.

The following image shows the assets that the Update Amazon Bedrock Knowledge Base recipe package contains:



Update Amazon Bedrock Knowledge Base recipe assets

The following table lists the assets that the Update Amazon Bedrock Knowledge Base recipe package contains:

Asset Name	Asset Type	Description
EmailConnectionBedrockUpdateDatasource	App connection	Email connection.
AmazonBedrockConnectionUpdateDatasource	App connection	Amazon Bedrock connection.
Send Email with Update Result	Process	Subprocess that sends an email with the process update result.
Update Amazon Bedrock Knowledge Base	Process	Process that accepts either an Amazon S3 bucket name to create a new data source and add all the files to the vector database, or a data source ID to directly add files to the vector database. If the files already exist in the database, the process updates the content. After all the files are added, an email notification is sent with the process results.

CHAPTER 2

Using the Update Amazon Bedrock Knowledge Base recipe

To use the Update Amazon Bedrock Knowledge Base recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Email connection.
3. Configure and publish the Amazon Bedrock connection.
4. Configure and publish the processes.
5. Invoke the process.

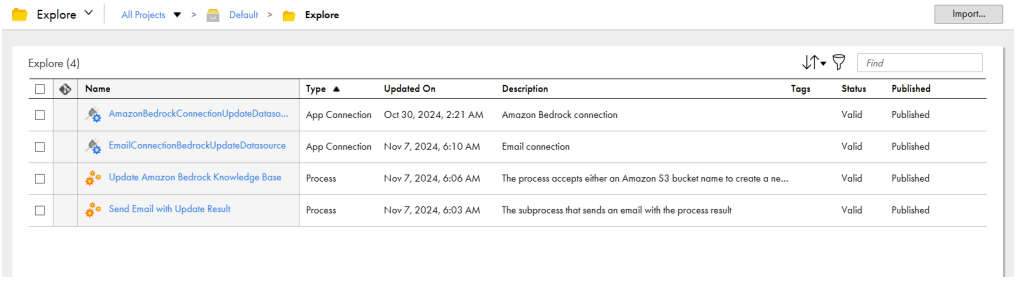
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Update Amazon Bedrock Knowledge Base** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Email connection

Configure the authentication type and authentication details in the EmailConnectionBedrockUpdateDatasource connection, and then publish the connection.

1. Open the **EmailConnectionBedrockUpdateDatasource** connection.
2. From the **Type** list, select **IICS Cloud Application Integration Email Service (Licensed for use)**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. From the **Authentication Type** list, select **Password** or **OAuth** as needed. Based on the authentication type selected, perform one of the following steps:
   - For **Password** authentication, enter values for the following properties in the **Connection Properties** section:

Property	Description
Authentication	Select **Enable**. Email Connector authenticates the user name and password that you enter in the email connection properties.
User Name	User name to log in to the email server. The user name is either the account name or the email address that is used to send the email with the synchronization results. For example: `notifyme@mydomain.com`
Password	Password for the email address.
Security	Select **SSL** for the Email connection to use the SSL protocol.

Configure the following common properties on the connection creation page:

Property	Description
Host	Email server's DNS name, such as `mail.mydomain.com`, or an IP address, such as `192.168.1.1`.
Port	Port for communication between the Process Server and the email server. Default is **25**.

- For **OAuth** authentication, enter values for the following properties in the **Connection Properties** section:

Property	Description
Authorization URL	Enter the OAuth authorization URL for the email service that is used to authorize the user request. For example: `https://login.microsoftonline.com/xxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx/oauth2/v2.0/authorize`
Token Request URL	Enter the OAuth token request URL that handles token requests. For example: `https://login.microsoftonline.com/xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx/oauth2/v2.0/token` The refresh token expires in 90 days. The user must authenticate again and publish the connection before the token expires.
Client ID	Specify the identifier value from the OAuth provider.
Client Secret	Enter the client secret to connect to the email application.
Scope	Specify the scope. The scope in OAuth authentication limits an application's access to a user's account. You can select multiple scopes for a single client. To enter multiple scopes, separate each value with a space. For a Microsoft Outlook email account, enter the following scope: `https://outlook.office.com/SMTP.Send offline_access`

Configure the following common properties on the connection creation page:

Property	Description
Host	Email server's DNS name, such as `mail.mydomain.com`, or an IP address, such as `192.168.1.1`.
Port	Port for communication between the Process Server and the email server. Default is **25**.

5. Save and publish the connection.

Step 3. Configure and publish the Amazon Bedrock connection

Configure the access key, secret key, and region to connect to Amazon Bedrock, and then publish the AmazonBedrockConnectionUpdateDatasource connection.

1. Open the **AmazonBedrockConnectionUpdateDatasource** connection.

2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
AccessKey	The access key to access Amazon Bedrock.
SecretKey	The secret key to access Amazon Bedrock.
Region	The region that hosts Amazon Bedrock.

3. Save and publish the connection.

Step 4. Configure and publish the processes

Configure the wait time to check the ingestion job status and publish the process.

1. To publish the **Send Email with Update Result** process, click **Actions** in the row that contains the process and select **Publish**.
2. Open the **Update Amazon Bedrock Knowledge Base** process.
3. In the **Set Flow Configuration** step, you can change the default wait time to check the ingestion job status in the **Waiting_Time_Ingestion_Job** field. Increasing the wait time will decrease calls to Amazon Bedrock.
4. Save and publish the process.

Step 5. Invoke the process

When you invoke the Update Amazon Bedrock Knowledge Base process, the process updates the existing Amazon Bedrock knowledge base using Amazon S3 data source and provide notification status.

1. Open the **Update Amazon Bedrock Knowledge Base** process and click **Actions** > **Properties Details** > **Copy Service URL**.
2. Open a text editor and add the input fields and values to the service URL as shown in the following format:

```
<Cloud Application Integration POD URL>/active-bpel/public/rt/<API_name>?
Knowledge_Base_ID=<Knowledge_Base_ID>&Email_Address=<Email_Address>&Bucket_Name=<Amaz
on_S3_Bucket_Name>&Data_Source_ID=<Data_Source_ID>
```

3. Open a browser and paste the service URL.

You can also invoke the process using the Run Using option, REST or SOAP API endpoints in any API client, such as cURL, Postman, SOAP UI, or through any programming language.